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Providing Support and Examples for Teaching Linear Equations in Secondary School: the Role of Knowledge of Mathematics Teaching

Ivonne Sandoval[1] · Montserrat García-Campos[1] · Leticia Sosa[2]

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Abstract

We present evidence of a secondary school teacher's *Knowledge of Mathematics Teaching* when she teaches linear equations in a word problem setting. We focus on analyzing the examples and mathematics support that she provides her students with during two classes. When faced with contingencies, she takes decisions whereby she adapts or creates emerging action plans in order to enrich and help her students overcome their difficulties to learn linear equations. The decisions and actions observed during the classroom practices demonstrated that providing support and examples for her students and accomplishing the lesson goals—mediated by her teaching knowledge—were at odds with each other.

Keywords Algebra teaching practices · Knowledge of Mathematics Teaching · Secondary school teachers · Examples and mathematics support · Mathematics teachers' classroom practices

The different ways in which the specialized knowledge of mathematics teachers appears are a recurring theme of interest in research (Stylianides & Stylianides, 2010; Scheiner et al., 2019; Zakaryan & Ribeiro, 2019). According to Delgado-Rebolledo and Zakaryan (2019, p. 4), "a new discussion of the nature of the mathematics teacher's knowledge allows us to better integrate knowledge and action". In view of the complexities involved in studying the knowledge *for* and *in* mathematics

Sadly, Dr García-Campos passed away on September 19th, 2021

Ivonne Sandoval
isandoval@upn.mx

Leticia Sosa
lsosa@uaz.edu.mx

1 Universidad Pedagógica Nacional, Mexico City, México

2 Benemérita Universidad Autónoma de Zacatecas, Zacatecas, México

teaching, there are different theoretical conceptualizations and approaches (e.g., Shulman, 1986; Rowland, 2008; Ball et al., 2008; Leikin & Zazkis, 2010; Carrillo et al., 2018). One aspect on which all of the foregoing agree is the relationship between the teacher's knowledge of the content to be taught and the pedagogical knowledge needed to transform that knowledge so that it becomes accessible to students (Shulman, 1986). That transformation can take place "both in planning to teach and in the act of teaching itself" (Rowland, 2014, p. 23).

The decisions made and actions taken by a teacher in her classes make it possible for students to draw closer to her specialized knowledge. In terms of Schoenfeld's (2000), actions *enacted* and decisions taken when teaching are framed by the teacher's knowledge, teaching goals, and beliefs. There may be different ways to achieve the goal; the teacher decides how to do so depending on several aspects such as her previous experiences, what happens in class, time, etc. The author points out,

> Goals may be pre-determined (e.g. as part of the lesson image) or they may be emergent (e.g. when the class seems restless, or an interesting issue arises in dialogue with the students) […] An action plan corresponding to a particular goal is a particular set of actions the teacher intends to undertake in order to achieve that goal. (p. 250–251)

In this study, our interest is to analyze the teacher's practice by way of her actions, and for this purpose, it is important to observe mathematics teachers' classroom practices[1] (Tirosh & Graeber, 2003) and provide an account of how the said understanding is put into action as the class unfolds (Sandoval et al., 2017). In particular, analyzing examples (Rowland, 2008; Zazkis & Leikin, 2007) and mathematics support that teachers give their students during class makes it possible for them to draw closer to their mathematics knowledge (Sosa et al., 2016).

Examples and support play an important role in mathematics teaching classroom practices. On the one hand, Zazkis and Leikin (2007) recognize the power of examples as a teacher' tool that provides a window into the mind of students. For those authors, "learning from examples" consists of explicit solutions to exercises shown by the instructor or in a text. Those examples are supposed to help because they show the use of different techniques that can be replicated or modified by students to solve similar exercises. In this case, the examples are provided by an authority (teacher or textbook), and it is the students who learn from them.

Meanwhile, Zodik and Zaslavsky (2008) state that examples and their representations are inseparable and are intended to help make mathematics understandable to learners. Said authors acknowledge that a core part of the teacher's work involves making decisions. In some cases, they can be anticipated with careful planning, while in others, they have to be made in real time so as to respond to unexpected

[1] We consider mathematics teaching classroom practices "along three dimensions: (1) providing opportunities for children to solve mathematical problems in their own ways, (2) listening to children's mathematical thinking, and (3) using children's mathematical thinking to make instructional decisions." (Tirosh & Graeber, 2003, p. 648).

situations that occur in the classroom (*contingency* according to Rowland (2008) in the Knowledge Quarter-KQ).

Choosing and generating examples generally requires making decisions on the spot in the classroom. A teacher's choice and use of examples is an action laden with responsibility that speaks to her flexibility and the necessary knowledge of mathematics to teach it (Zaslavsky, 2008). The examples can either help or hinder learning given that they constitute a fundamental communication scheme for explanations and mathematical discourse (op cit.).

There is little in the way of research on algebra knowledge for teaching in middle school. Among the existing research, McCrory et al. (2012) study different types of teacher knowledge at that school level, particularly teachers' knowledge of equations and how they used that knowledge to select and sequence tasks for their students. They suggest that there are "three key practices of algebra teaching—trimming, bridging, and decompressing—and the three domains of mathematical knowledge—school, advanced, and teaching" (p. 608). One of the three recommendations of the research focused on teaching algebra (Kieran, 2018; Star et al., 2015) suggests to "Use solved problems to engage students in analyzing algebraic reasoning and strategies." The foregoing accounts for "moving from a letters-symbolic and symbol-manipulation view to one that included multiple representations, realistic problem settings, and the use of technological tools" (Kieran, 2018, p. 3). In particular, research on the teaching and learning of linear equations is relevant to teaching, teacher training, and curriculum development (Castellanos et al., 2017; Depaepe et al., 2015; McCrory et al., 2012).

The objective of our research is to identify the *Knowledge of Mathematics Teaching* that is mobilized in the use of examples and mathematics support that a secondary school teacher provides students in the teaching of linear equations through problem-solving. In particular, we are interested in answering this question: How does a middle school teacher, with extensive teaching experience, choose, generate, and use examples and mathematics supports, in the sense of the MTSK framework, in order to support her students when they face difficulties to solve word problems of linear equations?

Theoretical Framework

The authors' view is that in teaching classroom practices, actions are carried out for specific objectives (framed in action plans), as is suggested by Schoenfeld (2000). Faced with student difficulties and errors, teachers deploy a series of actions so that the students can understand the topic and manage to overcome the difficulties.

To understand these actions in terms of the knowledge mobilized for teaching school algebra and for purposes of analysis, the authors of this paper use the model of *Mathematics Teachers Specialised Knowledge* (Carrillo et al., 2018) focusing on the role of the examples and mathematics support given to students (Sosa et al., 2016). The starting point for the model is the proposals of Shulman (1986) and the Mathematical Knowledge for Teaching (MKT) of Ball et al. (2008). However, in the MTSK, importance is attached to the meaning of "the specialized knowledge of mathematics teachers" seen as "a style of knowing that signifies specialization in mathematics teacher knowledge" (Scheiner et al., 2019, p. 153). We will examine mathematics teachers' knowledge in the sense of MTSK so as to contribute to the research that deals with that type of knowledge, relating it to classroom practices.

The model MTSK refers to the specific knowledge (as a whole) of the mathematics teacher and is composed of two domains, namely Mathematical Knowledge (MK) and Pedagogical Content Knowledge (PCK). Given the characteristics of the purpose of the research, linked to the teacher's practices, the focus of this work is essentially on the PCK dimension, in the sense of the MTSK framework (Carrillo et al., 2018). Below we describe the sub-domains that comprise it: Knowledge of Features of Learning Mathematics (KFLM) includes matters related to how certain concepts are learned; the intuitions, errors, and ways in which students interact with certain mathematics content. Knowledge of Mathematics Learning Standards (KMLS) refers to what can be expected (in terms of the national curriculum) at a given school level, also including the directions and means of learning such content from a "standardized benchmark" (Carrillo et al., 2018, p. 598), a benchmark provided by international organizations specializing in the didactics of mathematics.

For the data analysis, we will focus on the Knowledge of Mathematics Teaching (KMT). In this sub-domain, foremost are knowledge of teaching mathematics topics: knowledge of teaching theories (personal or institutionalized), knowledge of educational and manipulative materials, and knowledge of activities, examples, and mathematics support. As Sosa et al. (2016) state, KMT enables teachers to choose the most appropriate representations, the type of activities, and the technological resources to teach a given mathematics content. For instance, in our study (1) designing or adapting activities that make it possible to approach linear equations; (2) recognizing the potential of various resources and their use to solve the proposed activities; and (3) providing mathematics support or using examples with different intentions during management of classes. Table 1 shows the categorization previously developed by these authors.

In teaching, a teacher enacts a variety of actions related to her specialized knowledge and these actions speak to the complexity surrounding the work (Rowland, 2014; Scheiner et al., 2019). However, not all knowledge is acquired in institutions. Much of it is built in mathematics teaching classroom practices, in exchanges with other colleagues and in the very context where teaching is carried out. According to Leikin and Zazkis (2010), teacher knowledge for teaching mathematics is brought into action and can be enriched at the time classes are managed.

Methodology

In this work, our interest is in understanding the performance of a Mexican teacher in her teaching classroom practices. Therefore, we will analyze the actions that she carries out when she faces anticipated or unexpected situations during her teaching of linear equations. From an interpretative perspective (Denzin & Lincoln, 2005), we focus on the actions enacted by the teacher when she uses examples and mathematics supports in order for us to draw closer to her Knowledge for Mathematics

Table 1 Mathematics examples and aids (Sosa et al., 2016, p. X)

Category	Sub-category	Indicator
Activities, tasks, examples, and support	Knowledge of the features of tasks, examples, and activities	KMT1. Knowledge of the potential of contextual examples to create an environment of significance before introducing the formal definition of a concept.
		KMT2. Knowledge of the potential of examples as a means to highlight or emphasize the specific aspects of mathematics content that are meant to be taught.
		KMT3. Knowing how to create, choose, and explore problem situations in the form of examples or exercises, so that students can produce sense and meaning from the mathematics content.
	Knowledge of various ways of steering, channeling, and providing guidelines when faced with problems that arise in the learning of mathematics.	KMT4. Knowing what type of mathematics aids to give students in situations of error, confusion, or difficulty, for the purpose of enabling them to solve a task.
		KMT5. Knowing that a strategy for students to understand or carry out an example, exercise, or problem consists of pointing out what the activity requires of them.
		KMT6. Knowing how to point out specific problem data to students, data that is not explicit but that will be used to solve the problem.

Teaching and to comprehend the decisions taken in her classroom practices. We have focused on her as a case study (Bryman, 2016, p. 70) due to four of the teacher's traits: (1) teaching experience, with over 20 years of experience teaching high school she could be considered an expert teacher. In that sense, "The instructional practices most veteran teachers employ are fashioned to a large extent by their experiences in the classroom" (Guskey, 1989, p. 450); (2) interest in professional development to improve her teaching; (3) knowledge of the topic to be taught; and (4) use of different resources that were available to her (Adler, 2000). The teacher agreed to participate in this research voluntarily and also agreed to be videotaped (Sandoval et al., 2017). Earlier, she participated in a workshop (Garcia & Rojano, 2008) where she stood out for her mastery of curricular algebra content, her clarity when giving instructions to solve equations, and her knowledge of the use of graphing calculators (TI-89). Particularly, the teacher was able to coordinate the different representations for the algebraic objects offered by the calculator (tables and graphs). Furthermore, in this workshop, participants discussed algebraic activity (generational, transformational, global/meta-level) to classify tasks (Kieran, 2007), and she showed interest in implementing these new ideas among her students. As has been pointed out in other research "teachers who are most effective are also the most receptive to change and to the implementation of new ideas" (Guskey, 1989, p. 450).

In this vein, the intention differs from performing an evaluation; it is instead to draw the pedagogical content knowledge of one teacher that is being mobilized through her actions in teaching classroom practices closer. Rather, we are interested in understanding the complexity of behaviors and relationships that occur in the classroom, i.e. that set of mathematics actions, examples, and support provided by a teacher in view of the difficulties experienced by students in two linear equations classes. These classes were observed and videotaped, without direct researcher interference. One of the three researchers was present during the taping and had access to sheets of paper that container information on the specific problems designed by her for each class with the expected answers. We decided to focus our analysis using videotaped lessons for different reasons. On the one hand, doing so provided opportunities to recognize "complex behavioral patterns that would otherwise be difficult, if not impossible, to detect" (Savola, 2008, p. 180). On the other hand, videotaping lessons "is one of the most powerful tools for studying actual teaching behavior and classroom interactions" (Maulana et al., 2012, p. 838).

For the analysis, each of the researchers reviewed the videos, identifying goals that were materialized in an action plan, moments with examples and mathematics support in view of difficulties in the learning of linear equations (investigator triangulation). Groups then discussed observations, thus making it possible to identify key moments in teaching classroom practices. Each class was divided into episodes (Schoenfeld, 2000); each episode shows a sequence of actions related to the *action plan*. It should be noted that the episode is "a nested collection of sequences of action" that can be non-consecutive in time, i.e. that can be located at different points of time in the class. The action plans can be of three types: (a) predetermined (do not change); (b) they can be reconfigured with some familiar actions when faced with any instruction contingency (used in other previous experiences); or (c) they may arise on the spot (ad hoc attempts to deal with the contingency situation), which means that they can be partially structured in view of a new situation for the teacher (Schoenfeld, 2000).

At the time of structuring action plans and their implementation, beliefs and knowledge are brought into play. The teacher's actions in teaching practice depend on, among other things, pedagogical knowledge of the content.

The relationship between the teacher's goal (explicit or not) and the actions taken to achieve it at a specific time of her class is established in the analysis. For each episode, we built an objective (Specific Objective, SO) for actions that appear as a regular trace in the teaching practice classroom (Schoenfeld, 2000), whereas the Goal is explicitly expressed by the teacher. In the dialogs, C refers to the teacher, and A1, A2, ... , to students.

Data Analysis

In the classes observed, we identified that the teacher has a *predetermined action plan*, in which she plans to use different resources (graphing calculator, paper, and pencil). For each one she presents specific problems contained on a sheet of paper and she reads so that her students can write in their notebooks. Apparently, she drafted them for these classes. The problems can be used to show different techniques that students can replicate or modify (in the sense of Zazkis & Leikin, 2007).

Observing the implementation of an *action plans* enables us to identify (by way of evidence or indication) the specialized knowledge that is being mobilized. That is to say, some components of the KMT are interwoven, particularly those linked to examples and mathematics support to solve the difficulties that arise in the study of linear equations derived from word problems. We also identify specific and emerging objectives that enable one to understand the actions of the teacher during the classes.

The teacher allows the students to work in teams or individually. Her teaching classroom practices seem to follow this structure: starts by dictating a word problem, gives instructions on how to work in the class, and students work freely with or without a calculator. During the class, she walks around the classroom, clarifying any doubts (individual, in teams and groups), and reviews student productions. Table 2 shows the beginning of each class.

C's initial actions (see Table 2) lead us to assume that her pattern of action consists, on the one hand, of starting from a word problem that serves as an example, in order to create an environment of significance (KMT1 evidence; see Table 1). On the other hand, she intends to use different representations (verbal, geometric, table, algebraic, graphic) for the process of solving the problem, anticipating possible difficulties.

For the teacher, it is important to emphasize what she intends to teach, as is shown in class 1 (KMT2 evidence). Apparently, when she specifies the purpose of class by emphasizing "Solve the equation for the unknown," she seems to recognize potential difficulties that students will face in solving an equation. This decision may be the result of C's teaching experience. In the workshop, she expressed that the

Table 2 Beginning of both classes: word problems

Class 1	Class 2
C: (The teacher dictates from her worksheet) **Topic**: **Linear equations.** General purpose: **Solve the equation for the unknown** in different types of formulas. Relate a formula to the table of data generated and its graph. Problem. In a store 9 packages of books are bought, which has **an additional price of** $75 [sic]. How much does each package cost if the total paid was $1,400? C: Read the problem. Once you've stated it, you can use the calculator to solve the problem. You can solve it individually, in twos or threes, depending on how many students are sitting at each table. [Fragment 1, class 1, 1:52–4:00]	(In the student notebooks, one can see the following statement: The length of the room in the king's house is **twice** as long as its width. If its perimeter is 21 m, what are its dimensions?) C: I want to see the problem statement [Fragment 1, class 2, 39:00–39:40] (The teacher dictates another problem) C: I'm going to dictate a second problem. You're going to solve by dragging the pencil or using the calculator, whichever you prefer. She dictates the following: The perimeter of a rectangular piece of land is 56 meters. If the length **is three times** the width, what are the dimensions of the piece of land? [Fragment 2, class 2, 58:00–59:49]

main difficulty for her students has to do with algebraic syntax (using pencil and paper and CAS) when solving word problems.

From the teaching classroom practice, in her *predetermined plan* for class 2, the teacher incorporates a geometric context that students are to generate as of the word problem, then state and solve the equation, and check the result. It seems that the problem contexts become increasingly complex (i.e. first use of the term "double" and then "three times"), and she introduced a different representation in addition to the previous lesson (geometric representation) to expand the level of significance (we infer these actions as an indication of KMT1). As we will show throughout episodes 1 and 2, we also identified that the gradual nature is also seen in interactions with and the support given to her students. That is to say, class 1 was more guided, while for the second problem in class 2 she gives students greater autonomy. Our interpretation is that she uses this first problem of class 2 as an example, and expects the students to replicate the equation solution techniques (Zazkis & Leikin, 2007) in the next problem.

Below, we present two episodes in which the teacher's actions in both classes converge and make it possible to identify examples and mathematics support. Such actions are linked to implemented action plans (predetermined, adapted, or emerging). It should be noted that these episodes have specific objectives. For each episode, the actions are evidenced at different times of each class and are recognized as fragments (sub-episode, Schoenfeld, 2000). Table 3 shows the episodes and their corresponding specific objectives.

E1. Understanding the Statement and Stating the Equation

This episode speaks to the actions, which are directly related to the *action plans*, undertaken by the teacher to maintain the objective of the classes in view of a contingency. She undertakes the actions at different key moments for her and throughout

Table 3 Description of the episode and its corresponding specific objective

Name of the episodes (E)	Specific objective (SO)
E1. Understanding the statement and stating the equation.	SO1. Properly interpret the information given in the word problem (data, relationship between data and question) in order to generate a linear equation that solves it.
E2. Comparison of procedures and verification of results.	SO2. Compare different solutions (obtained via different procedures, using different resources) of the equation to identify the correct and appropriate solution to the problem.

Table 4 Translation of the statement into its equation

Class 1	Class 2
C: You have to **read your problem**. **Once you've stated the problem**, you can use the calculator to solve it [...] C: Your classmate says that when **the term *additional* is used it refers to [...]** A: That you have to **increase**. C: In this case, how much do you have to **increase**? Student: 75 C: What would your equation look like? [...] (Addresses the entire group) Have you read it, **all of the text**? What do you mean? **What does the problem consist of?** You're going to draw a table (addresses a student) Based on what are you going to make the table? [Fragment 2, class 1, 00–06:50]	C: **How do we represent the double?** A: **Squared** C: **Double** A: **Two x** (C sends a student to the blackboard) C: ... **just write the equation that you're going to work with for them** [Fragment 3, class 2, 44:53–45:30] C: Why did you **represent it in that way**? A: Because the x is the width, since we don't know what it is (C interrupts the student's explanation) C: [...] **Draw them a rectangle.** Write the data for them there. If that helps you to tell us (referring to explaining the context of the statement). […] C: **Because that's how to calculate the perimeter of what** (pause) A: a rectangle C: Why a rectangle and not a square? [Fragment 4, class 2, 46:17–50:33]

the entire class. Based on the analysis of the data presented in Table 4, we identified two moments when C needs to change her predetermined action plan regarding general purpose (see Table 2, class 1): in the first, where one single term generates different interpretations of the problem, and second, representation of the unknown and its relationship with the context of the problem. In this section there are fragments from both classes. Highlighted in bold are the expressions analyzed.

Table 4 shows how interpretation of the word *additional* (class 1) in the wording of the statement leads the teacher to change her predetermined action plan. Different interpretations lead some teams to consider different procedures to solve the problem and to different responses (see episode 2). However, the teacher requests students to return to the statement text, perhaps as a way to help (indication of KMT4) so that they are able to state the equation contemplated in her *predetermined action plan*. The intent of this decision would seem to create an environment of significance of the word problem (assumption of KMT1). Students spontaneously begin to explore with numerical values (a solution that Kieran, 2006, call *substitution by trial and error*). These actions are, however, insufficient to solve the contingency presented (see fragment 3 discussion about interpretation of the word *additional*). At this time a new objective emerges, which we infer from the teacher's actions, "unify the interpretation of the term *additional* in the context of the problem" so as to state the equation.

Filloy et al. (2008, p. 18) claim that the word *additional* is not identified with the mathematical action of adding, rather with real actions such as lengthening or expanding. In this case, the students tend to use meanings derived from natural language, when what they need is to afford this sense prior to symbolizing it. This diversity of responses from students attests to the inconsistency of interpretation, which results in an incorrect symbolization.

The same applies to class 2 with the term *double* that creates a situation that seems familiar to the teacher, confusing *double* (2x) with *square* (x^2). In order to expand the level of significance related to the problem statement and its symbolization, her action is to emphasize the geometric interpretation. That is, one side (width) is double the other (height) in a figure (example is prototype of rectangle) and therefore cannot be a square. In addition, students must evoke the meaning of perimeter and how is it calculated, information that is not explicit in the word problem. Mathematics support from the teacher aims at pointing out that data to the students, data that is essential to stating the equation (KMT6 evidence). Filloy, et al., (2008, p. 18) state that some students tend to interpret expressions such as *double* or *square* exclusively in terms of geometric formulas.

C's actions allow us to identify three types of examples and support in translation of the statement to algebraic language in both classes and is evidence of knowledge.

- Indication of KMT1. The words *additional* and *double* have specific meanings in algebra, which determines a particular way to symbolize them. The correct interpretation of the statement requires that *additional* be translated as *increase* or *add* (see Table 4, fragment 2, class 1) and *double* as *twice* (see Table 4, fragment 3, class 2). The teacher uses them as an example to create an environment of significance by reading the statement and posing questions for reflection at different times during the class
- Evidence of KMT4, in order to clarify the confusion generated by interpretation of the term *additional* and its relationship with the problem data, and the error of considering *double* as *square*. As mathematics support, she suggests that students first focus on stating an equation that models the problem, before carrying out operations (see Table 4, fragment 2, class 1; fragment 4, class 2). That is to say, she proposes a strategy so that students are able to understand the statement and can translate it into symbolic algebraic
- Evidence of KMT6. The meaning of perimeter is required for use of the geometric model (see Table 4, fragment 4, class 2), as is the difference between square

Table 5 Stating the equation

Class 1	Class 2
C: **What does your problem want to know?** It wants to know how much a package costs, how many packages were bought? A2 and A3: Nine. [...] A 4: You subtract seventy-five from the latter. C: [...] Again, a package cannot cost you seventy-five pesos. **It is additional. Let's consider it as an increase, after you know how much the price of a package is.** A5: 80 (refers to a package price) C: **But not a package, rather the total.** But how would you state it? [...] What would your equation look like? **What does our topic say?** (the students read what the teacher dictated to them at the beginning) **We need to state the equation using logic.** The equation that arises from a problem, not just any equation. A4: Teacher, does the 1,400 already include the 75 or not yet? C: (She addresses student 4). **You won't be able to state it until you understand it.** [She addresses the entire group.] **I need to see the equation that will solve your problem.** Many are already tabulating. What are you going to tabulate? I only want to see how you state it. [Fragment 3, class 1, 8:40–11:31]	C: [...] **Read your problem, what does it want?** What does your problem say? A: (Reads the problem out loud.) C: **To state the problem, we need to read it well and understand it.** If it were a square, what would it say? A: The sides of a square are equal, and one side of a rectangle is wider than the other. C: That's why is was a rectangle and not a square. Now **you have the likely equation** there, now solve it using the calculator. If you don't remember how to use the calculator, just you drag the pencil. [Fragment 5, class 2, 46:17–50:33] C: But remember that **there are two equal sides.** That condition must be satisfied (pointing to the writing in the student notebook) **that the length is twice the width.** C: If you tell me that one side measures 8. How much will the width measure if you tell me that the length measures 4, if it **is double?** [Fragment 6, class 2, 50:40–52:00]

and rectangle. As a means of mathematics support, she points out problem data that do not appear explicitly to her students

Regarding the statement of the equation and its algebraic symbolization as part of episode E1, changes concerning the predetermined action plan are identified. For class 1, an emerging plan and in class 2, an adaptation, with a common goal that "all students are able to state one same equation." Her strategy is to highlight and establish relationships between some of the numerical values related to the word problem. In the case of class 1, for instance, the 9 that corresponds to the number of packages, and the 75 linked to the additional cost. In class 2, it is to have students establish the relationship between width and length of the rectangular room, and to produce the equation that represents the perimeter. Table 5 presents a dialog of the exchange between the students and the teacher.

Table 5 (end of fragment 3, class 1, and fragment 5, class 2) shows how C's actions emphasize asking her students to find the equation that represents each problem, i.e. what the task requires (KMT5 expression). From her teaching classroom practice, once the equation has been stated, students can do calculations, find numerical values, and build the corresponding variation table, an inverted process compared to what they used to do—exploring with numerical values (trial and error) to state the equation (fragment 3, class 1). The teacher apparently knows what the correct answer is from her *action plan*. However, interpretation of the term *additional* leads the students to at least two different solutions. Once the students propose their solutions (displayed in episode E2), C takes advantage of this situation to invite then to yet again read the problem and review the equation statement.

In class 2, the teacher establishes the difference between the sides of a square and any non-square rectangle in order to state the equation. She intends to have her students write an expression in which, on the left side of the equal sign they have operations to be carried out with the unknown and, on the right side, the value of the perimeter included in the statement. With those examples, students can make sense of the geometric representation, to state the corresponding equation (KMT3 evidence).

In both classes, her intention is that her students identify the unknown, the relationship between the data and the unknown, and use of the equal sign. These actions relate to achieving the objective that guides her class, both in the predetermined plan of action and in the adaptations undertaken. However, in class 1, making changes to her predetermined plan was not enough; she had to improvise an emerging plan. The dialog below shows the contingency situation derived from the different interpretations of the term *additional* in the wording of the word problem.

> A6: The thing is that **the problem does not state if it's 75 for each one or for all of them.** And here we were able to prove that it was 75 for all of them. Because it does not work out for it to be for each one, that way it would be 1447.95
> C: [...] (she addresses the entire group). **Read it again.** What is it that isn't clear?
> A6: If it's 75 for each package or if it's an additional 75 for the 9 packages. **The statement is missing a word.**
> C: [...] **What word would it be? For it to be clear to us.**
> A6: It would have to say **that its an additional 75 pesos for each package** [...]
> C: That's the purpose of problems [...] Each student will solve it differently [...] What do they ask you? **How much does each package cost, but not each book.**
> A8: (He reads). 9 packages of books, then it says, **which has** an additional price of 75 pesos.
> C: To the cost of the 9 packages, an additional 75.
> A6: [...] It's not just reading, but **determining the relations between the data and solution.**
> [Fragment 4, class 1, 33:40–39:00]

The previous dialog is evidence of the exchange among students in which they try to identify what causes the differences between the interpretations of the same statement and, therefore, different responses (these two procedures will be shown at E2). C asks her students why there are different interpretations in order to provide the mathematics support needed for them to solve the problem (KMT5 evidence). Apparently, she believes that it consists of a difficulty in understanding the statement, and insists that they read it again. However, A6 tells her that "The sentence is missing a word. [...] that it's 75 additional pesos for each package." During the exchange of arguments, C listens to her students and notes that they interpret the expression "which" as referring to "each package," "all packages," or "each book." At that point, her actions focus on having her students deduce and, together, identify what "which" means, given that it is not explicit in the statement (KMT6 evidence).

The teacher recognizes that the wording of the statement is unclear, and a new objective emerges, which is to "clearly express in the statement the relations among the data of the problem." Her actions are evidence of her ability to adapt to unforeseen situations, and she makes the most of the situation to highlight the connection between the statement, its interpretation, stating the equation, and its solution.

In episode E1, the changes in the predetermined plan of actions are evidenced, as well as how a contingency in class 1 (see Table 5) prompts the teacher to generate an emerging plan that questions her predetermined action plan.

Her actions to ensure that her students state the equation in each problem enable us to identify knowledge with the following examples and mathematics support.

- Evidence of KMT3. The interpretation of *double* relates to the perimeter formula of a rectangle, but not a square. She explores, by way of an example, the difference between the sides of these quadrilaterals so that the students can make sense and produce meaning of the mathematics content (see Table 5, class 2)
- Evidence of KMT4. The word *additional* implies the appropriate use of parentheses in the equation. In view of the confusion about whether it is to all or to each package, as a means of mathematics support, she requests her students to read the wording of the problem for the purpose of having them solve the task (see Table 5, class 1)
- Evidence of KMT5. Interpretation of the word problem determines establishment of relations among data in an equation. In order to decide which of the different interpretations of the statement is correct (fragment 4, class 1), as mathematics support, she proposes stating them out loud to the students. In this way, they may achieve what the activity requires of them, that is to state the equation
- Evidence of KMT6. The expression "which" is a determining factor in establishing the relationship between the data and the equation: "each package," "all packages," or "each book." When she identifies that her students interpreted the expression in different ways (fragment 4, class 1), her mathematics support consists of making explicit the data from the problem that it is crucial to solving the task

E2. Comparison of Procedures and Verification of Results

In this episode, the teacher keeps her predetermined action plan prompting, by way of mathematics support, her students to compare different solutions to one seemingly same equation so as to identify the solution that is correct and appropriate for the problem. Teacher C may be using some ideas from the workshop on algebraic meaning building (Kieran, 2007). To achieve this, we identify two actions related to mathematics support: one, making comments out loud (for the whole group) to student questions concerning the procedures performed, and another, by sending students that used the correct procedure to the blackboard (which is what she planned), as well as simultaneously have students present the procedures or results. She uses some of her students' attempts to contrast errors and procedures, and to make clarifications for the entire group. These actions are intended to help students solve the equation correctly. To better understand the teacher's actions, in this episode, we initially examine the mathematics support in class 1 where students solve the problem using two procedures (see Tables 6 and 7), then later, the mathematics support in class 2, where the problem is solved by way of the same procedure using different resources (see Table 8).

Class 1. Arithmetic and Algebraic Procedures

C identifies two procedures. The first requires operations with numbers (known or unknown), and the second requires operating with the unknown.

The following dialog shows how C achieves the first objective of her predetermined action plan (OE1), stating the equation.

C: Look at what your classmate is going to do (addressing the entire group)
A9: Writes on the blackboard: $9x + 75 = 1400$.

Table 6 Confrontation of two different procedures

Arithmetic procedure	Algebraic procedure
$75 \times 9 = 675$ (additional)	$9\times + 75–75 = 1400–75$
$1400.00 - 675.00 = 0725.00$	$9x = 1325$ $\frac{9x}{9} = \frac{1325}{9}$
$\frac{725}{9} = 80.25$(1 leftover)	$x = 147.22$
[22:37–23:36]	[20:10–20:14]

Table 7 Verification of results of each procedure

Arithmetic procedure	Algebraic procedure
$80.55 \times 9 = 724.95$	$147.22 \times 9 + 75 = 1400$
$724.95 + 725.00 = 1447.95$	(As a result of the discussion between both students, one arrives at a:)
	$(147.22 \times 9) + 75 = 1400$

C agreed with the conclusion of the two students and expresses it to the entire group:

It doesn't matter if you made a mistake, here it is a matter of finding out how to solve it. You can solve it in this way, and she can do it in that other way [...] It **is a matter of seeing which of the two procedures is correct or appropriate at a given time.** [Fragment 7, class 1, 32:25–33:20]

Table 8 Comparison of solutions to the same equation

Blackboard	Calculator
A12:	A11:
$4x + 4 = 21$	solve $(x + (2x) + x + (2x) = 21, x)$
$4x = 21 - 4$	$x = 7/2$
$4x = 17$	$x = 3.5$
17/4 (performs the division)	
$x = 4.25$	
C: They're saying that there is a mistake, let him continue checking it.	[Fragment 7, class 2, 56:45–57:00]
C: Tell your classmate where the mistake is.	
A13:	
$6x = 21$	
$x = 3.5$	
length $2x$= 7 m	
width 3.5 m	
	[Fragment 8, class 2, 58:00–58:16]

> C: [...] We wanted **to arrive at this, and now how can we solve that equation.**
> [Fragment 5, class 1, 13:49–15:35]

Now, their attention is on the solution process to achieve the second objective (OE2), comparison of different solutions to one same equation in order to identify the correct solution. The teacher shows evidence of KMT3, as she chooses two procedures that provide different results, by way of example, so that her students can build sense from solving an equation. Table 6 shows the procedures written by two students.

With these examples, C does not identify why they produce two different results. It may perhaps be because the unknown does not appear in the arithmetic procedure, and she cannot compare the results. So at this point, she does not know how to support her students. Her strategy is to ask them to justify their procedures and results, as shown in the following dialog.

> C: I would ask the young lady to explain why she solved it that way, and the other girl to come forward and explain it [as well]. Let's try **together to find out which of the two procedures would be correct to arrive at the answer.**
> (Each student explains her procedure).
> C: **Assuming both procedures are correct, what would the results be?**
> Group: The same
> C: That means that one of the two is incorrect.
> [Fragment 6, class 1, 27:38–30:33]

The mathematics support this example highlights is KMT4, given that when faced with the confusion, she asked the two students at the blackboard to verify the results. In this way, she was able to have the students notice that the difference between the procedures lay in the use of parenthesis, and its relationship with the interpretation of the term *additional*. In this case, the corresponding equations would be: for the arithmetic procedure $9(x+75) = 1400$ and for the algebraic procedure $(9x) + 75 = 1400$. We identified student difficulties concerning two respects (Filloy et al., 2008): the polysemy of the x and interpretation of the term *additional*, which was expressed in how students (implicitly) used the parenthesis.

Table 7 shows how the verification of results made it possible to conclude that the correct procedure is algebraic.

In the fragment above, to explore the problem the teacher contrasts the two procedures and the difference in the answers, by way of an example, so that the students can produce meaning from the relationship between the word problem, the way the equation is stated and the result. She emphasizes that even if the procedures are different, the answer has to match (evidence of KMT3).

C's actions, aimed at having her students solve the problem, allow us to identify examples and mathematics support, and evidence of her knowledge.

- Evidence of KMT3. In the environment of word problems, equations and results must be consistent regardless of the procedure used. As an example, she leverages the result verification process to significantly analyze the relationship (see Table 7)
- Evidence of KMT4. In an equation, use of parentheses determines relations among data and, therefore, has an impact on the result. Given two different procedures and answers on the blackboard (see Table 6), as a means of providing support, she confronts them simultaneously and asks the students to check the results so as to identify the corresponding equation

Class 2. Comparison of the Procedure Using Different Resources

The word problem to be solved, as mentioned in E1, is: "The length of the room in the king's house is twice as long as its width. If its perimeter is 21 m, what are its dimensions?" The students used different syntax (see Table 8) to express the relationships among the data in the problem. It was clear that they had difficulty performing the operations to the left of the equal sign, but they know that the right side is 21—the value of the perimeter.

C identified that they used two resources (calculator and paper and pencil) to solve the equation. She decides to send one student per resource to the blackboard in order to provide a group example. With these actions, her intention is to do away with the confusion related to the algebraic syntax (expression of KMT4). When comparing the results, the responses from the two students are different. The teacher identifies the mistake made by A12, and that A11 solved the problem correctly. The error corresponds to difficulties with some conventions of algebraic notation (Filloy et al., 2008), in particular, symbolic conventions of transformations in the solution of equations.

As an example, C uses the pencil and paper procedure performed by another student, A13, for the purpose of having the group see the algebraic expression that corresponds to the problem. Table 8 shows the attempts of the three students.

Another of C's actions, as can be seen in fragment 9, is to promote results verification with the intent of having her students confirm that the value of x is correct.

A11: (Writes 2 ways of checking on the calculator)

3.5+ 3.5*2+3.5+3.5*2	3.5+7+3.5+7
21	21
	[Fragment 9, class 2, 54:50–56:41]

The teacher explains the student's procedure to check that, in the first case, the student used 3.5 to substitute the value of x and, in the second, that the student multiplied 3.5 by 2 to get the result of 7. She uses verification as an example to show the importance of substitution of a literal value in the formula for the perimeter and the use of symbolic conventions of operations and transformations in the solution of equations, i.e. to highlight singular aspects of the solution of a linear equation (KMT2 evidence).

In addition, she leverages the example of A13's production so that her students can produce sense and meaning from the unknown (width) and its relationship with the length (KMT3 evidence). At the same time, she points out what the activity requires within a word problem setting (KMT5 evidence), obtaining the dimensions (length and width) and their relationship to obtain the perimeter of the rectangular room (see fragments 8 and 9).

C's actions in comparing the procedures and checking the results enables us to identify examples and mathematics support, and evidence of mathematical knowledge.

- Evidence of KMT2. The substitution of the value of the literal is used to validate the solution of an equation. As an example, she leverages result verification to highlight unique aspects, such as the use of symbolic conventions of operations and their transformations (fragment 9).
- Evidence of KMT3. The unknown and its relationship with the data of the problem in an equation determine the result. As an example, she leverages the verification of results (using the two resources) to build meaning in the relationship between the statement of the problem, the unknown and the result (see Table 8, fragments 8 and 9).
- Evidence of KMT4. In an equation, the concatenation of terms affects the result. Given two procedures (with two resources) and different answers (see Table 8), by way of mathematics support, she confronts them simultaneously so that they are able to find the correct value of the literal.
- Evidence of KMT5. When the value of the unknown in an equation is found, one must return to the context of the word problem. By way of mathematics support, she uses A13's productions to point out what the activity requires, that is, to obtain the dimensions of the rectangular room (see Table 8, fragment 8).

Discussion

From the analysis of the proposed activities (three word problems), the examples, and mathematics support that the teacher gave her students in two classes observed, we are able to recognize knowledge related to teaching linear equations. That knowledge is expressed in situations for (1) translation of the problem wording into algebraic language, (2) stating the equation, (3) solving the equation, and (4) checking the results.

In the analysis, we identified teacher actions that respond to the needs of the students, the majority of which related to difficulties concerning how to state and solve linear equations in the context of word problems (Kieran, 2018). Her actions are representative of how she uses Knowledge of Mathematics Teaching.

The actions identified focus on examples and mathematics support. The examples, in the sense of Zazkis and Leikin (2007), are given by the teacher, or the students if they have teacher validation (in her role of authority). We identified that the examples are generally intended to make the mathematics content comprehensible, and they are shown by way of (1) different procedures (arithmetic/algebraic), (2) solution techniques (grouping of similar terms and substituting the value of the unknown) so that they can be replicated by students when solving other similar problems, and (3) a specific word problem to solve others that are more complex. The foregoing results coincide with those of Rowland (2014, p. 24) with novice teachers, in the sense of "the use of examples to assist concept formation and demonstrate procedure." However, with the examples chosen and used, this teacher was consistent with the intended pedagogical purpose of her *action plan*.

The mathematics support was generally given to solve difficult[ies] translating from natural language to algebra and vice versa, which have been reported in other studies (Filloy et al., 2008), in particular terms, with difficulties related to concatenation of terms, conventions of algebraic notation, and being unable to formally express the methods and procedures for solving problems. Such means of support had different intentions:

1. Read the statement so as to (a) build sense and meaning for terms such as additional (add / increase), double (square vs. twice), triple, and square figure vs rectangular figure; (b) understand the statement and translate it into algebraic language; and (c) provide a solution to the task.
2. Compare and justify procedures so as to (a) find the value of the literal and (b) find the equation that corresponds to the statement.
3. Verify results to (a) identify the correct equation, (b) analyze the relationship among statement-equation-result, (c) build sense and meaning from statement-equation-result, and (d) highlight the importance of substituting the value of the literal in the formula (perimeter of a rectangle).

C's examples and mathematics support provide "opportunities for children to solve mathematical problems in their own ways" (Tirosh & Graeber, <yearelement class="findWord" contenteditable="true" id="6">2003</yearelement>,

p. 648). The students propose different procedures (correct/incorrect; arithmetic/algebraic, using paper and pencil/calculator) and C leverages that to enrich the class and delve into aspects related to solving linear equations.

These examples and means of mathematics support were managed by C either on an individual basis, working with teams, or groups of students. She used individual support to generate group support. The goal of this mathematics support is to have students arrive at the correct answer, and she generally provides it by way of questions. In this case, although the teacher focuses her examples and mathematics support on one student or team of students, she speaks out loud so that the entire group can hear and participate in the discussion. These actions show how C uses "listening to children's mathematical thinking" (Tirosh & Graeber, 2003, p. 648) by promoting discussion and explanation of student ideas.

The two episodes provided speak to actions implemented and knowledge for teaching linear equations, in terms of examples and mathematics support, provided by the teacher in view of the difficulties, mentioned above, experienced by her students. Such actions enabled identification of mathematics knowledge for teaching, in terms of:

- *Features of tasks, examples, and activities* (KMT1, KMT2, KMT3). The emphasis was on having her students produce sense and meaning from the mathematics content by way of the examples.
- *Different ways of directing, channeling, and giving guidance in view of the problems that arise in the learning of mathematics* (KMT4, KMT5, KMT6). The emphasis of the mathematics support was to solve the task in situations of mistakes or confusion, pointing out to the students any data that did not explicitly appear in the word problem.

Mathematics support and examples provided in the two classes are evidence that the teacher intended to maintain the predetermined objective: state and solve linear equations. However, C's reaction in contingency situations depends on what (e.g. drafting of the statement, interpretation, equations) or who raises them (the students or the teacher herself).

In class 1, the contingency comes from the wording of the statement (contingency caused by her), i.e. the task itself. We identified Knowledge of Mathematics Teaching expressed in generating and implementing an emerging plan through actions such as promoting a reading of the statement (episode 1, class 1) and checking results (episode 2, class 1). Her actions enable her to leverage the contingency in order to delve into aspects linked to the sense and meaning of words in natural language within an algebraic context. This emerging plan focused on mathematics support, first for herself (find out why there were two results) and then for her students (deciding between two different interpretations, both semantic and syntactically to obtain the correct equation). That flexibility and adaptation of C's to the contingencies in her classes speak to how she "us[es] children's mathematical thinking to make instructional decisions" (Tirosh & Graeber, 2003, p. 648).

In class 2, the contingency comes from the three equations that have different syntax (see Table 8) raised by the students, and not the difficulties in interpreting the statement. C adapts her predetermined plan and uses her Knowledge of Mathematics Teaching so that her students have the correct procedure (default order), by promoting verification of results. Such actions highlight what Bozkurt and Ruthven (2017, p. 325) assert as "the way in which experience-based craft knowledge accumulates, as a teacher becomes familiar with an increasing range of classroom situations and develops effective ways of managing them".

The teacher's reaction to the contingencies identified in the two classes was that they represented an opportunity for "opening the door for exploration" (Rowland & Zazkis, 2013, p. 150) and as "teacher insight during instruction" (Rowland, 2014, p. 25), expressed in the form of examples and/or mathematics supports. We infer that is a teacher with the confidence to negotiate and make sense of mathematical situations' relationship of solving problem with linear equation.

The methodological design of our research (observation of two classes) does not allow us to identify the reasons justifying her decision-making and actions in the classroom. However, when analyzing C's teaching classroom practices (class 2), we noted the existence of opportunities from the Mexican teacher training programs, for enriching the discussion, for instance, using the three equations with different syntax (see Table 8) to create emerging objectives or adapt them as examples or mathematics support in order to delve deeper into other topics of school algebra. In particular, it is possible to use the potentiality of A12's mistake, $4x + 4 = 21$ as a result of operating $x + 2x + x + 2x = 21$, a mistake related to "concatenation of terms," that is, operate with an arithmetic benchmark; on one side, the coefficients (numbers) and on the other, the literal (x). Also, the teacher could have made leveraged A11's calculator equation, $x + (2x) + x + (2x) = 21$, and contrasted it with A13's, $6x = 21$. The first corresponds to an interpretation of the perimeter of the rectangular room, side+side+side+side = 21, and not the formula to calculate the perimeter of rectangles. Such situations might be used as an example to (a) emphasize aspects related to operating algebraic expressions, (b) discuss whether they are "equivalent" expressions (A11's and A13's), (c) analyze three different translations of natural to algebraic language, and (d) return to the context of the problem, in terms of giving meaning and sense to "double" and "rectangular." The latter could for example use the answer obtained by A13 of "length $2x = 7$m" and "width 3.5," and return to the question posed in the problem, what are the dimensions of the rectangular room?

Discussions on the teaching and learning of linear equations are still underway in the Mathematics Education community (Kieran, 2018). Not only because of trying to understand the complexities of the subject within school algebra, but also because of the implications for teaching based on analysis and reflection on teaching classroom practices.

Final Remarks

Teacher knowledge for teaching mathematics is brought into action and can be enriched at the time classes are managed. Leikin and Zazkis (2010, p. 18) emphasize that "Teachers learn both mathematics and pedagogy when teaching." In the case of teacher C, her knowledge seems to be enriched when contingency situations arise. The authors reach this conclusion because in our analysis, we identified how to promote the discussion of two different interpretations of the word "additional" and make them explicit, namely (1) *additional per pack*, i.e. $9(x + 75)$, and (2) *additional per nine packs*, $9x + 75$. Such situations generate challenges for teachers since the decisions and actions enacted during teaching classroom practice are framed by their knowledge, *goals* (*predetermined action plan*) and beliefs (Schoenfeld, 2000). In this vein, C's *predetermined action plan* is compromised (jeopardized) and she needs to change it (class 2) or generate an *emerging action plan* (class 1). This is one of the contributions of the article—provide evidence for approaching an understanding of relationships between "what knowledge teachers have and how it is manifest in practice, a need" pointed out by authors such as McCrory et al. (2012, p. 585) "the measures of effectiveness say nothing about how teachers use their knowledge in practice."

Observing C's two classes revealed the need to promote reflection on and leveraging student mistakes or situations of confusion (Nathan & Koedinger, 2000; Rowland, 2008). Such situations are valuable, and we believe they can be explored and incorporated into action plans (predetermined, adapted or emerging) by way of examples or mathematics support so as to create sense and meaning related to the concepts to be taught.

These results represent an approach to studying the teaching of algebra in middle school, and they represent progress on the first recommendation proposed by Star et al. (2015) for teaching algebra: "Use solved problems to engage students in analyzing algebraic reasoning and strategies". Nonetheless, more research is needed to address that first recommendation with other contents, as well as to address the other two recommendations: "2. Teach students to utilize the structure of algebraic representations" and "3. Teach students to intentionally choose from alternative algebraic strategies when solving problems." (Kieran, 2018).

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References

Adler, J. (2000). Conceptualising resources as a theme for teacher education. *Journal of Mathematics Teacher Education, 3*(3), 205–224.

Ball, D. L., Thames, M. H., & Phelps, G. (2008). Content knowledge for teaching: What makes it special? *Journal of Teacher Education, 59*(5), 389–407. https://doi.org/10.1177/0022487108324554

Bozkurt, G., & Ruthven, K. (2017). Classroom-based professional expertise: a mathematics teacher's practice with technology. *Educational Studies in Mathematics, 94*(3), 309–328. https://doi.org/10.1007/s10649-016-9732-5

Bryman, A. (2016). *Social research methods*. Oxford University Press.

Carrillo, J., Climent, N., Montes, M., Contreras, L., Flores-Medrano, E., Escudero-Ávila, D., Vasco, D., Rojas, N., Flores, P., Aguilar-González, A., Ribeiro, M., & Muñoz-Catalán, M. (2018). The mathematics teacher's specialised knowledge (MTSK) model*. *Research in Mathematics Education, 20*(3), 236–253. https://doi.org/10.1080/14794802.2018.1479981

Castellanos, M. T., Flores, P., & Moreno, A. (2017). Reflections on future mathematics teachers about professional issues related to the teaching of school algebra. *Bolema: Boletim de Educação Matemática, 31*(57), 408–429. https://doi.org/10.1590/1980-4415v31n57a20

Delgado-Rebolledo, R., & Zakaryan, D. (2019). Relationships Between the Knowledge of Practices in Mathematics and the Pedagogical Content Knowledge of a Mathematics Lecturer. *International Journal of Mathematical Education in Science and Technology*, 1–21. https://doi.org/10.1007/s10763-019-09977-0

Denzin, N. K., & Lincoln, Y. S. (2005). Introduction: The discipline and practice of qualitative research. In N. K. Denzin & Y. S. Lincoln (Eds.), *Handbook of qualitative research* (pp. 1–32). Sage.

Depaepe, F., Torbeyns, J., Vermeersch, N., Janssens, D., Janssen, R., Kelchtermans, G., & Van Dooren, W. (2015). Teachers' content and pedagogical content knowledge on rational numbers: A comparison of prospective elementary and lower secondary school teachers. *Teaching and Teacher Education, 47*, 82–92. https://doi.org/10.1016/j.tate.2014.12.009

Filloy, E., Rojano, T., & Puig, L. (2008). *Educational algebra: A theoretical and empirical approach* (43rd ed.). Springer Science & Business Media.

García-Campos, M., & Rojano, T. (2008). Appropriation processes of CAS: A multi dimensional study with secondary school mathematics teachers. In O. Figueras, J.L. Cortina, S. Alatorre, T. Rojano & A. Sepúlveda, (Eds.), *Proceedings of the Joint Meeting of PME 32 and PME-NA XXX* (Vol. 1, p. 260). Cinvestav-UMSNH.

Guskey, T. R. (1989). Attitude and perceptual change in teachers. *International Journal of Educational Research, 13*(4), 439–453. https://doi.org/10.1016/0883-0355(89)90039-6

Kieran, C. (2006). Research on the learning and teaching of algebra: A broadening of sources of meaning. In A. Gutiérrez & P. Boero (Eds.), *Handbook of Research on the Psychology of Mathematics Education: Past, Present and Future* (pp. 11–49). Brill, Sense Publishers.

Kieran, C. (2007). Learning and teaching algebra at the middle school through college levels. Building meaning for symbols and their manipulation. In F. Lester (Ed.), *Second Handbook of Research on Mathematics Teaching and Learning* (2nd ed., pp. 707–762). InformationAge Publishing, Inc. and NCTM.

Kieran, C. (2018). Algebra Teaching and Learning. In S. Lerman (Ed.), *Encyclopedia of Mathematics Education*. Springer. https://doi.org/10.1007/978-3-319-77487-9_6-5

Leikin, R., & Zazkis, R. (2010). Teachers' opportunities to learn mathematics through teaching. In R. Leikin & R. Zazkis (Eds.), *Learning Through Teaching Mathematics: Mathematics Teacher Education* (5th ed., pp. 3–21). Springer. https://doi.org/10.1007/978-90-481-3990-3_1

Maulana, R., Opdenakker, M. C., Stroet, K., & Bosker, R. (2012). Observed lesson structure during the first year of secondary education: Exploration of change and link with academic engagement. *Teaching and Teacher Education, 28*(6), 835–850.

McCrory, R., Floden, R., Ferrini-Mundy, J., Reckase, M. D., & Senk, S. L. (2012). Knowledge of algebra for teaching: A framework of knowledge and practices. *Journal for Research in Mathematics Education, 43*(5), 584–615. https://doi.org/10.5951/resematheduc.43.5.0584

Nathan, M. J., & Koedinger, K. R. (2000). Teachers' and researchers' beliefs about the development of algebraic reasoning. *Journal for Research in Mathematics Education, 31*, 168–190.

Rowland, T. (2008). The purpose, design and use of examples in the teaching of elementary mathematics. *Educational Studies in Mathematics, 69*(2), 149–163. https://doi.org/10.1007/s10649-008-9148-y

Rowland, T. (2014). The Knowledge Quartet: The genesis and application of a framework for analysing mathematics teaching and deepening teachers' mathematics knowledge. *SISYPHUS Journal of Education, 1*(3), 15–43.

Rowland, T., & Zazkis, R. (2013). Contingency in the mathematics classroom: Opportunities taken and opportunities missed. *Canadian Journal of Science, Mathematics, and Technology Education, 13*(2), 137–153. https://doi.org/10.1080/14926156.2013.784825

Sandoval, I., Solares, A., & García-Campos, M. (2017). Teaching trategies and mathematics teacher's specialized knowledge. A case in School Algebra. In: E. Galindo & J. Newton, (Eds.), *Proceedings of the 39th annual meeting of the north american chapter of the international group for the psychology of mathematics education* (pp. 567–580). Hoosier Association of Mathematics Teacher Educators.

Savola, L. T. (2008). *Video-based analysis of mathematics classroom practice: Examples from Finland and Iceland*. [Unpublished doctoral dissertation]. Columbia University.

Scheiner, T., Montes, M., Godino, J., Carrillo, J., & Pino-Fan, L. (2019). What makes mathematics teacher knowledge specialized? Offering alternative views. *International Journal of Science and Mathematics Education, 17*(1), 153–172.

Schoenfeld, A. H. (2000). Models of the teaching process. *The Journal of Mathematical Behavior, 18*(3), 243–261. https://doi.org/10.1016/S0732-3123(99)00031-0

Shulman, L. S. (1986). Knowledge and teaching: Foundations of the new reform. *Harvard Educational Review, 57*(1), 1–22.

Sosa, L., Flores-Medrano, E., & Carrillo, J. (2016). Conocimiento de la enseñanza de las matemáticas del profesor cuando ejemplifica y ayuda en clase de álgebra lineal. *Educación matemática [Mathematics Education Review], 28*(2), 151–174.

Star, J. R., Caronongan, P., Foegen, A., Furgeson, J., Keating, B., Larson, M. R., Lyskawa, J., McCallum, W. G., Porath, J., & Zbiek, R. M. (2015). Teaching strategies for improving algebra knowledge in middle and high school students. In *National Center for Education Evaluation and Regional Assistance (NCEE)/Institute of Education Sciences/U.S*. Department of Education.

Stylianides, G. J., & Stylianides, A. J. (2010). Mathematics for teaching: A form of applied mathematics. *Teaching and Teacher Education, 26*(2), 161–172. https://doi.org/10.1016/j.tate.2009.03.022

Tirosh, D., & Graeber, A. O. (2003). Challenging and Changing Mathematics Teaching Classroom Practices. In A. J. Bishop, M. A. Clements, C. Keitel, J. Kilpatrick, & F. Leung (Eds.), *Second International Handbook of Mathematics Education. Springer International Handbooks of Edu cation* (10th ed., pp. 643–687). Springer.

Zakaryan, D., & Ribeiro, M. (2019). Mathematics teachers' specialized knowledge: A secondary teacher's knowledge of rational numbers. *Research in Mathematics Education*, 21(1), 25–42. https://doi.org/10.1080/14794802.2018.1525422

Zaslavsky, O. (2008). What knowledge is involved in choosing and generating useful instructional examples? In *Paper presented at the Symposium on the Occasion of the 100th Anniversary of ICMI*. ICMI.

Zazkis, R., & Leikin, R. (2007). Generating examples: From pedagogical tool to a research tool. *For the Learning of Mathematics, 27*(2), 15–21. https://doi.org/10.1007/s11858-011-0334-5

Zodik, I., & Zaslavsky, O. (2008). Characteristics of teachers' choice of examples in and for the mathematics classroom. *Educational Studies in Mathematics, 69*(2), 165–182. https://doi.org/10.1007/s10649-008-9140-6